Akshay Meshiya

Validation Engineer
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PROFESSIONAL SUMMARY

Engineer with an M.S. in Chemical Engineering from Rowan University and 6+ years in regulated manufacturing, quality control, and process engineering, now pursuing validation engineering. I have commissioned and tested equipment against specifications, written test documentation and SOPs, investigated deviations with root cause analysis (RCA), and closed them with CAPA under change control (MOC). I apply statistical process control (SPC), data analysis, and good documentation practices (GDP), and I have working knowledge of commissioning and qualification (C&Q), IQ/OQ/PQ protocols, and risk-based validation.

WORK EXPERIENCE

Process Engineer **Apr 2026 – Present**
Valcor Engineering Corporation **NJ, USA**

- Support $2.5M+ in CAPEX projects by commissioning ultra-high-purity (UHP) gas and liquid delivery systems and linking them to SAP PP/PM for maintenance visibility and equipment traceability.
- Lead HAZOP and LOPA studies with root cause investigations, identifying 10+ safety and operational risks and keeping the site audit-ready under strict regulatory requirements.
- Reduce process variability in batch and continuous operations with statistical process control (SPC) and downtime analysis, raising throughput while holding cleanroom-grade purity.
- Build and maintain OSIsoft PI System and Power BI dashboards that track critical KPIs and process trends in real time, speeding up fault detection and cutting unplanned downtime by 15%.
- Develop PFDs, P&IDs, and 3D plant layouts in AutoCAD Plant 3D and SmartPlant P&ID for 12 equipment upgrades that raised unit production capacity by 10%, managed through management of change (MOC).
- Model high-purity chemical processes in Aspen Plus, Aspen HYSYS, and ChemCAD, improving solvent recovery efficiency by 7% while meeting environmental compliance standards.

Assistant Production Engineer **Feb 2022 – Jun 2024**
R.R. Chemical (ISO 9001:2015 Certified) **Gujarat, India**

- Ran quality control (QC) tests to confirm product integrity and specifications under ISO 9001.
- Investigated batch deviations with root cause analysis (RCA, 5 Whys) and closed them with corrective and preventive actions (CAPA), reducing batch failures by 15%.
- Wrote and updated SOPs, MSDS/SDS, and EHS documents to keep procedures inspection-ready.
- Supported scale-up and technology transfer from R&D to manufacturing, recording process parameters, procedures, and results.
- Kept experiment and batch records accurate using good documentation practices.
- Fixed process and equipment problems on reactors, pumps, and process lines and verified performance after each fix.

Process Executive & Chemist **Jul 2018 – Feb 2022**
Palsana Enviro Protection Ltd. (PEPL) — Common Effluent Treatment Plant **Gujarat, India**

- Tested inlet, in-process, and treated effluent for pH, COD, BOD, TSS, TDS, color, oil & grease, MLSS, and SVI.
- Checked treated effluent against GPCB/CPCB discharge norms and helped prepare compliance reports and records for regulatory inspections.
- Kept shift logbooks, lab records, and daily MIS reports using good documentation practices, and escalated deviations to plant management.
- Replaced handwritten logbooks with Microsoft Excel trackers and trend charts for lab results, chemical use, and flow, which cut shift reporting and MIS preparation time by 30%.
- Used jar tests to fine-tune coagulant, flocculant, lime, and polyelectrolyte doses, cutting chemical consumption by 8% and lowering treatment cost per unit of effluent.
- Watched data trends to catch process deviations early, which helped operations avoid downtime and off-spec discharge.
- Ran shift operations at a common effluent treatment plant (CETP) handling wastewater from textile units, across primary, secondary, and tertiary treatment.
- Sampled effluent from member industries to catch non-compliant discharges before they upset downstream treatment.

SKILLS

- **Validation & Quality:** Commissioning & Qualification (C&Q), IQ/OQ/PQ Protocols (working knowledge), Test Documentation, Good Documentation Practices (GDP), Deviation Investigation, CAPA, Change Control, ISO 9001, Quality Control, Audit Readiness
- **Risk & Compliance:** Risk Assessment, HAZOP, LOPA, Root Cause Analysis (RCA), Management of Change (MOC), SOP Development, OSHA Process Safety Management (PSM), EHS, Regulatory Compliance
- **Data & Statistics:** Statistical Process Control (SPC), Data Analysis, Trend Analysis, Minitab, Python (Pandas, Matplotlib), Microsoft Excel (Advanced)
- **Process Engineering:** PFD/P&ID Development, Process Optimization, Scale-Up, Technology Transfer, Material & Energy Balances, Process Control (PID Loops, DCS/PLC Fundamentals)
- **Software:** SAP (PP/PM), OSIsoft PI System, Power BI, AutoCAD, AutoCAD Plant 3D, SmartPlant P&ID, Aspen Plus, Aspen HYSYS, ChemCAD, Microsoft Office

PROJECTS

Manufacturing of Ethylamine — B.E. Final Year Design Project **2021**

- Designed a conceptual 100 TPD ethylamine production facility; performed material and energy balances, equipment sizing, reactor and distillation design, cost estimation, and process safety evaluation.

Manufacturing of Ethylene Glycol — Diploma Final Year Design Project **2018**

- Executed process design, feasibility analysis, material balance calculations, utility assessment, and process economics evaluation.

EDUCATION

Master of Science, Chemical Engineering — Rowan University, Glassboro, NJ **Aug 2024 – May 2026**

Relevant Coursework: Process Optimization, Advanced Reactor Design, Transport Phenomena, Advanced Engineering Thermodynamics, Polymer Engineering, Water & Environment Monitoring, Sustainability Assessment, Strategic Engineering Management

Bachelor of Engineering, Chemical Engineering — Gujarat Technological University **Jul 2018 – Jul 2021**

Relevant Coursework: Process Equipment Design I & II, Mass Transfer Operations I & II, Chemical Reaction Engineering I & II, Process Heat Transfer, Fluid Flow Operations, Chemical Engineering Thermodynamics I & II, Instrumentation & Process Control, Process Modeling, Simulation & Optimization, Multicomponent Distillation, Plant Design & Project Engineering, Pollution Control, Safety & Health Management

Diploma, Chemical Engineering — Gujarat Technological University **Jul 2015 – Jun 2018**

Relevant Coursework: Mechanical Operations, Process Heat Transfer, Mass Transfer, Industrial Stoichiometry, Chemical Process Technology I & II, Utilities & Instrumentation in Chemical Plant, Pollution Control & Effluent Treatment, Safety & Hazard Management in Chemical Industry, Petroleum Refining & Petrochemical Technology